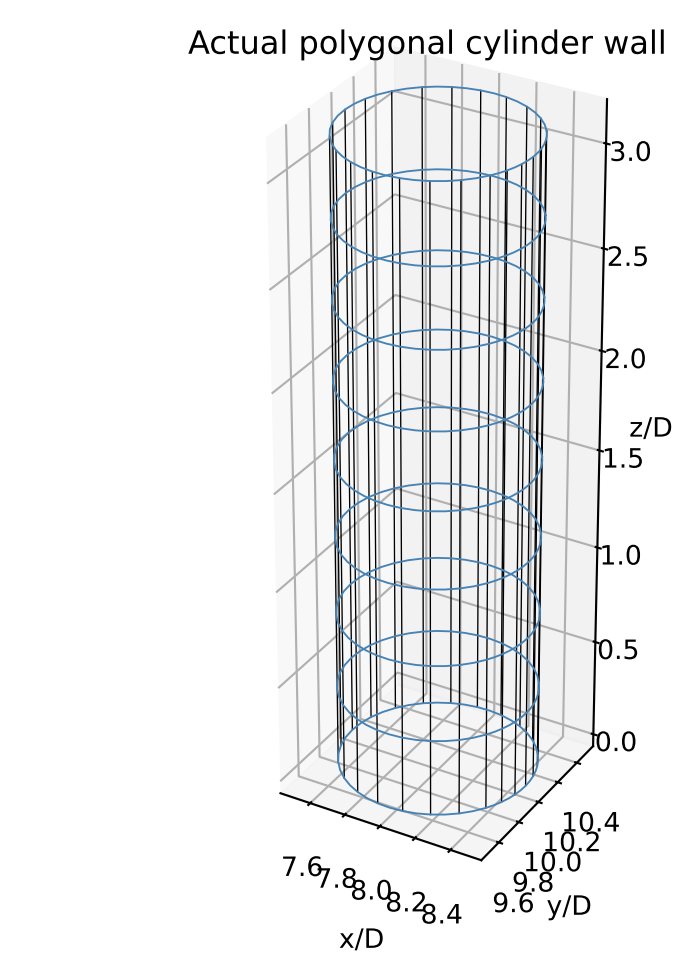
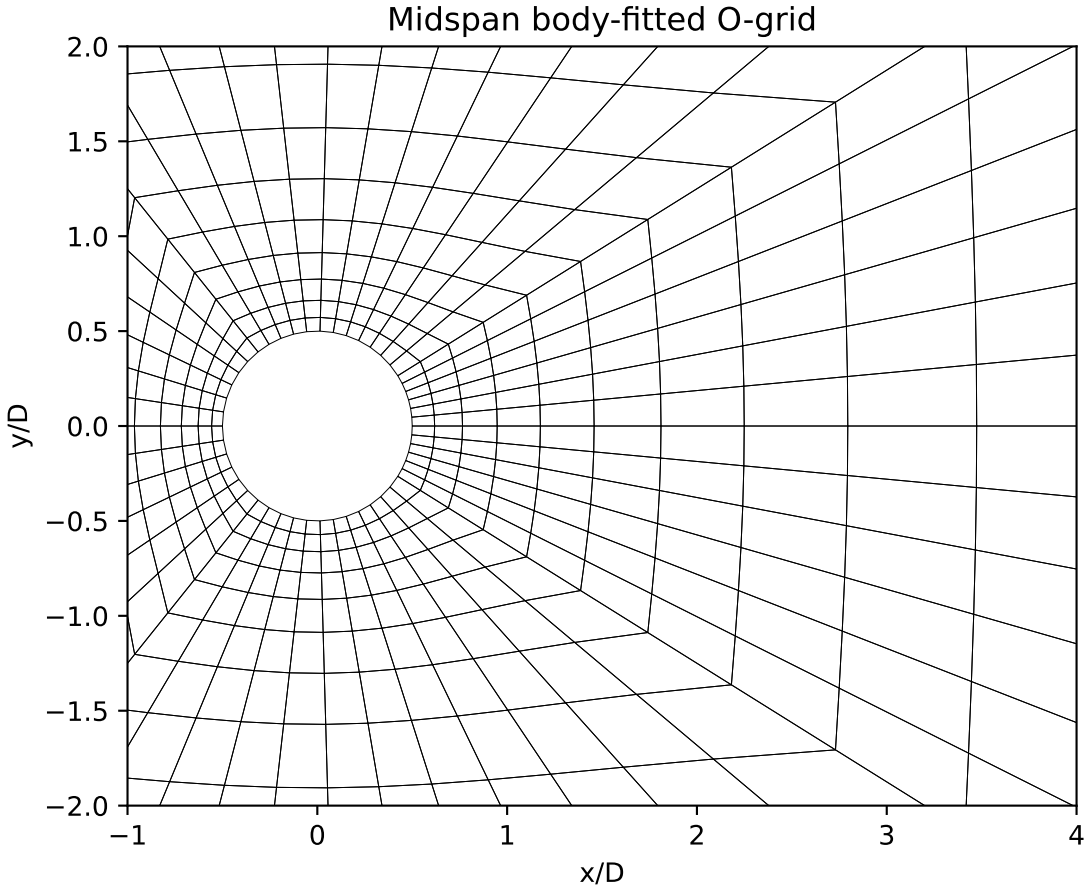
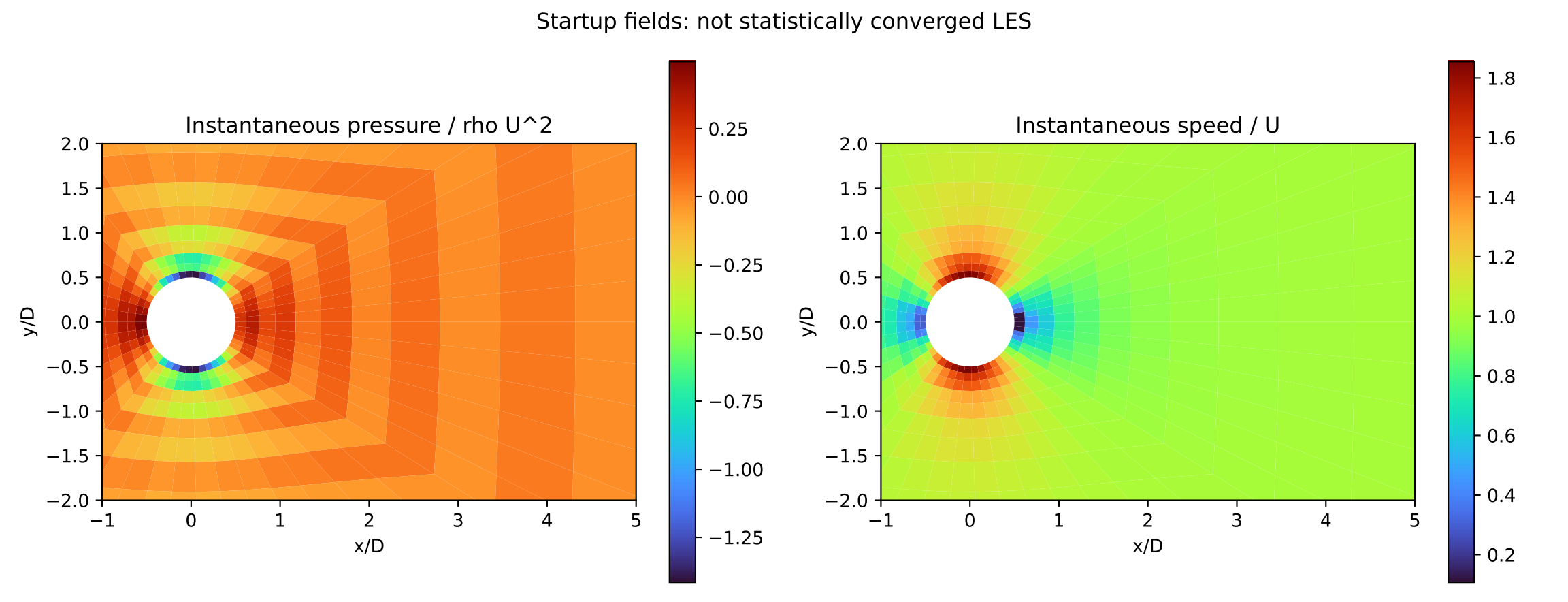
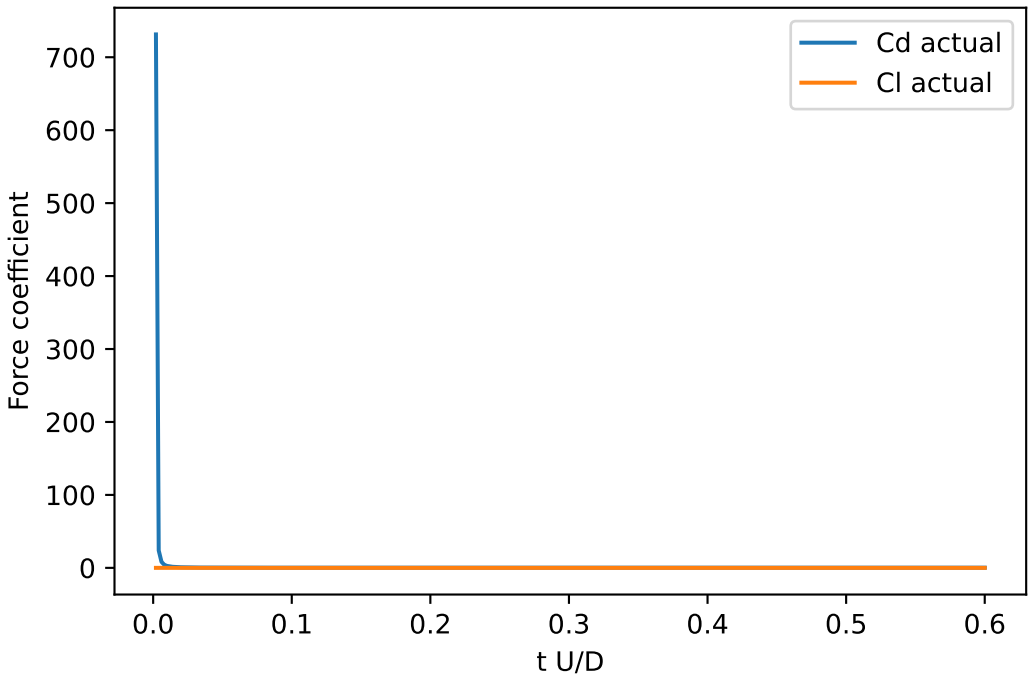
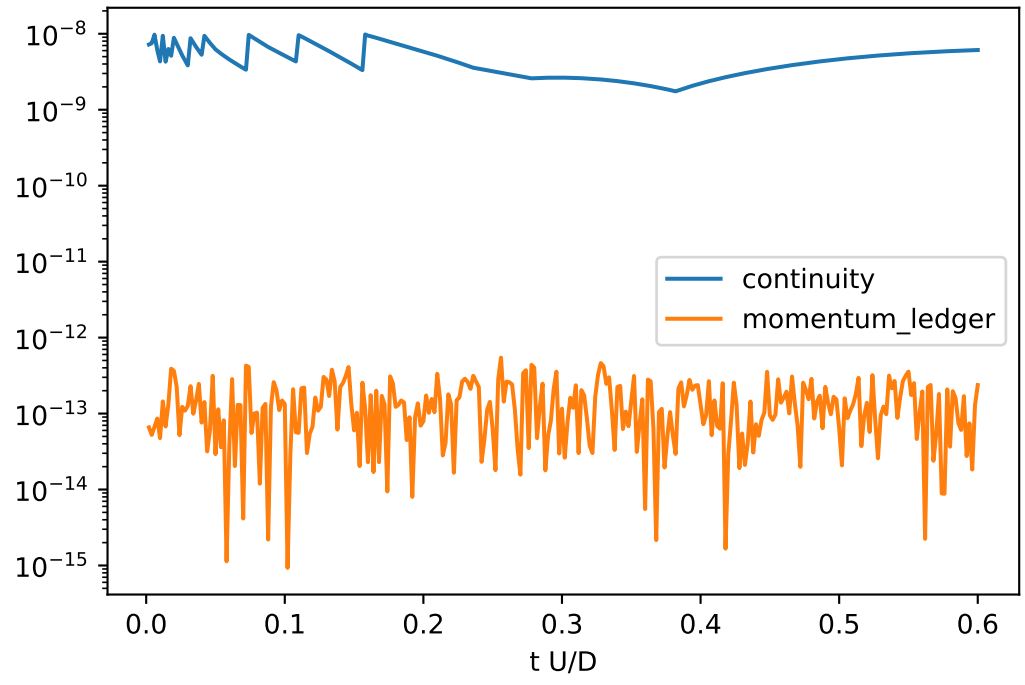
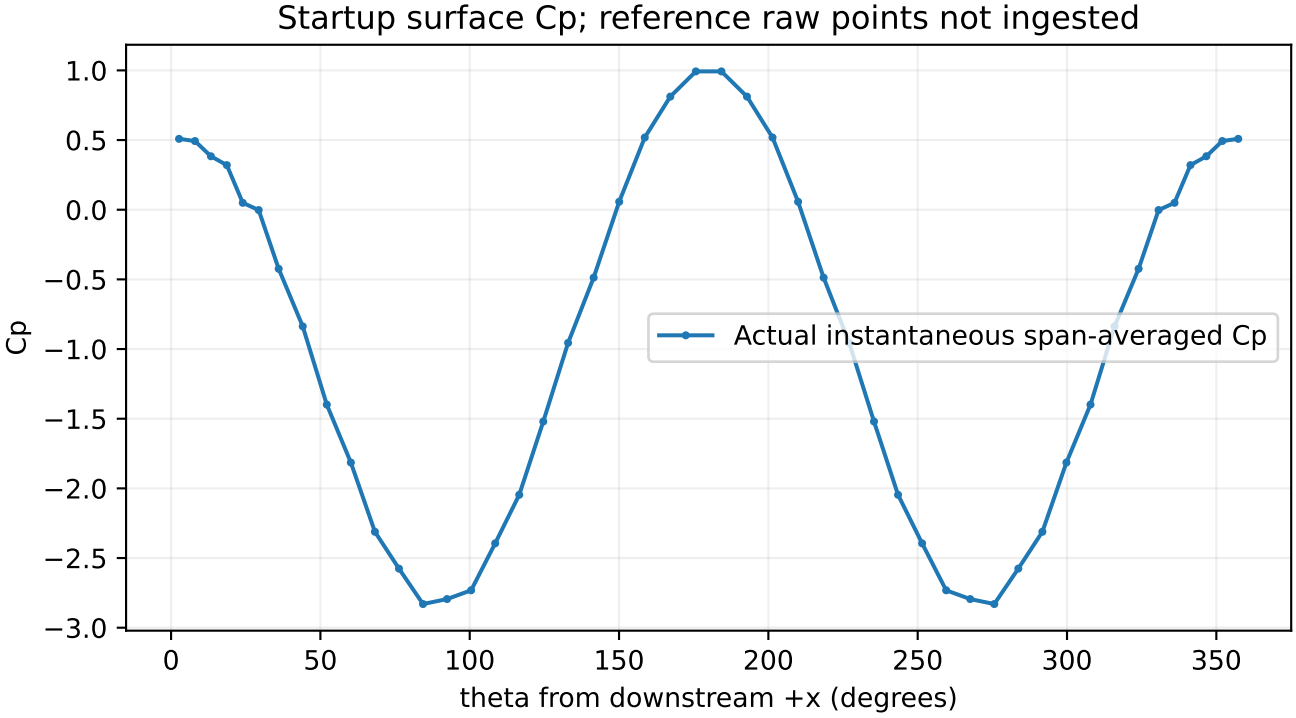

Actual polygonal cylinder wall
3.0
2.5
2.0
z/D
1.5
1.0
0.5
0.0
10.4
10.2
10.0
9.8
9.6
y/D
7.6
7.8
8.0
8.2
8.4
x/D

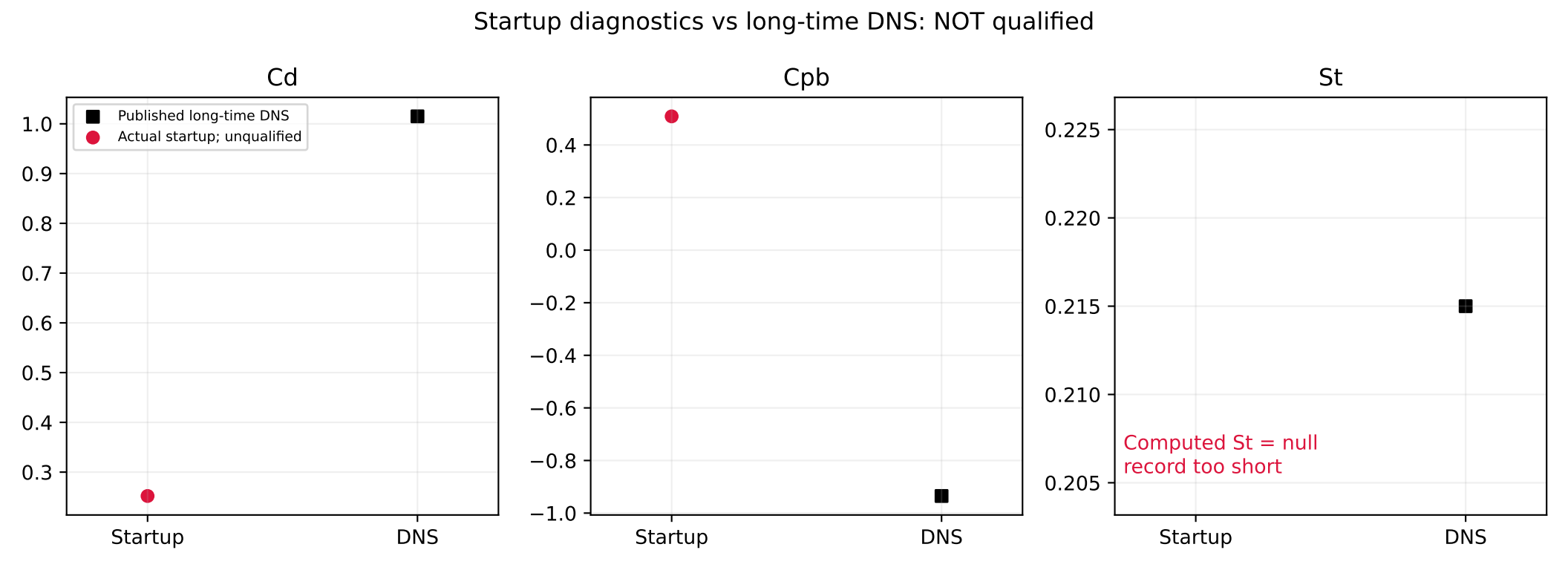

Midspan body-fitted O-grid
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
y/D
−1
0
1
2
3
4
x/D


Startup fields: not statistically converged LES
Instantaneous pressure / rho U^2
Instantaneous speed / U
y/D
x/D
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1
0
1
2
3
4
5
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2


Startup force, not time-averaged reference
Cd actual
Cl actual
Force coefficient
700
600
500
400
300
200
100
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
t U/D


Independent mass and momentum ledgers
$10^{-8}$
$10^{-9}$
$10^{-10}$
$10^{-11}$
$10^{-12}$
$10^{-13}$
$10^{-14}$
$10^{-15}$
continuity
momentum_ledger
0.0
0.1
0.2
0.3
0.4
0.5
0.6
t U/D


Startup surface Cp; reference raw points not ingested
Cp
theta from downstream +x (degrees)
Actual instantaneous span-averaged Cp
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
0
50
100
150
200
250
300
350


Startup diagnostics vs long-time DNS: NOT qualified
Cd
Cpb
St
Published long-time DNS
Actual startup; unqualified
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0.225
0.220
0.215
0.210
0.205
Computed St = null
record too short
Startup
DNS
Startup
DNS
Startup
DNS